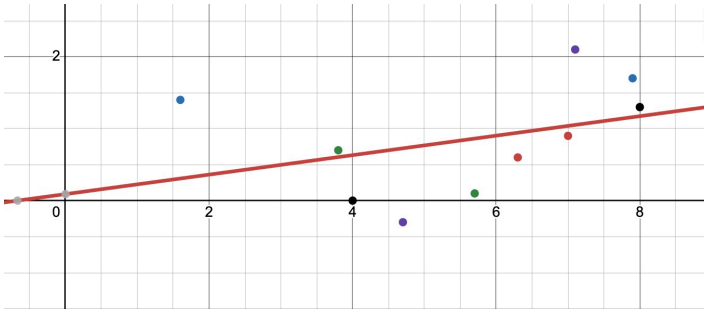
Q1

(a) $\sum x = 56.1 \quad \sum y = 8.5 \quad \sum x^2 = 353.09 \quad \sum xy = 52.88$

$\theta_1 = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} \approx 0.1354 \quad \theta_0 = \bar{y} - \theta_1 \bar{x} = 0.0904$

$\therefore \hat{f}_\theta(x) = 0.0904 + 0.1354x$

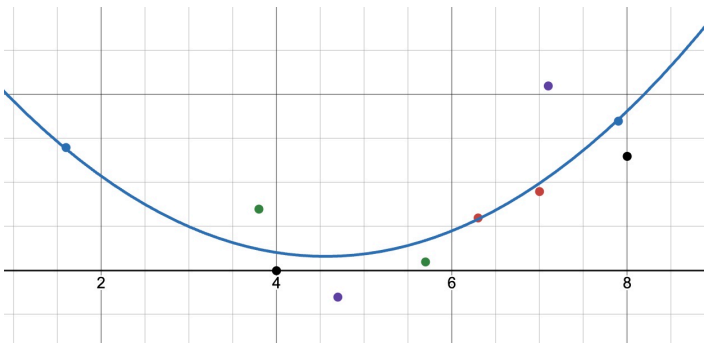


(b) $J(\theta) = \|X\theta - Y\|_2^2 = (X\theta - Y)^T(X\theta - Y)$

$\frac{\partial J}{\partial \theta} = 2X^T(X\theta - Y)$ let $\frac{\partial J}{\partial \theta} = 0 \Rightarrow X^TX\theta = X^TY \Rightarrow \begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \sum x_i^2 y_i \end{bmatrix}$

Solve the equation $\Rightarrow \theta_0 = 3.06 \quad \theta_1 \approx -1.271 \quad \theta_2 \approx 0.1394$

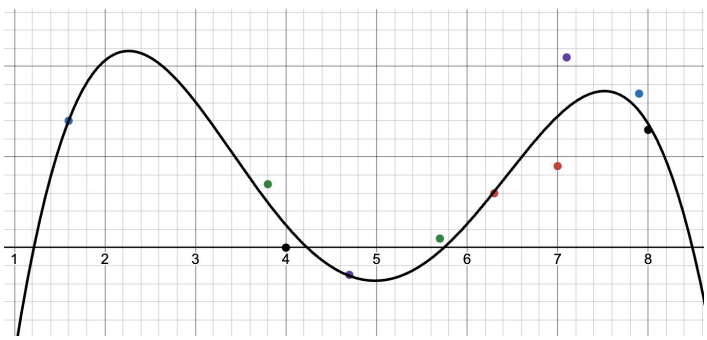
$\therefore \hat{f}_\theta(x) = 3.06 - 1.271x + 0.1394x^2$



(c) Use Python to calculate $\Rightarrow$

$\theta_0 \approx -12.1769 \quad \theta_1 \approx 16.4458 \quad \theta_2 \approx -6.3777 \quad \theta_3 \approx 0.9547 \quad \theta_4 \approx -0.0485$

$\therefore \hat{f}_\theta(x) = -12.1769 + 16.4458x - 6.3777x^2 + 0.9547x^3 - 0.0485x^4$



Simply use $\sum(\hat{y}_i - y_i)^2$ to determine the error

After calculation using the data in Test Set

RSS / Model	Train	Test
Linear	4.782	4.12
square	1.877	2.477
Quartic	0.771	5.184

Linear: Small Rss diff in 2 sets but both to high $\Rightarrow$ underfitting

Square: relatively small diff and small Rss in 2 sets $\Rightarrow$ good

Quartic: two big diff in 2 sets' RSS, low robustness $\Rightarrow$ overfitting

(d) underfitting: $\hat{f}_\theta(x) = \theta_0 + \theta_1 x$

relatively good: $\hat{f}_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2$

overfitting: $\hat{f}_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

Q2

(a) let $\frac{\partial J}{\partial \theta_0} = 2\sum_{i=1}^{m}\left[x^{(i)}\theta + \theta_0 - y^{(i)} \right] = 0 \Rightarrow \sum_{i=1}^{m} x^{(i)}\theta + m\theta_0 - \sum_{i=1}^{m} y^{(i)} = 0$

$\therefore m\theta_0 = \sum_{i=1}^{m} y^{(i)} - \sum_{i=1}^{m} x^{(i)}\theta \Rightarrow \theta_0 = \bar{y} - \bar{x}\theta$

$\because \frac{\partial J}{\partial \theta_0} = 2m > 0 \quad \therefore$ when $\theta_0 = \bar{y} - \bar{x}\theta$, we can get the min value for $J(\theta, \theta_0)$

(b) After pluging, $J(\theta, \theta_0) = \sum_{i=1}^{m}\left[x^{(i)}\theta + \bar{y} - \bar{x}\theta - y^{(i)} \right]^2 + \lambda\|\theta\|_2^2$

$= \sum_{i=1}^{m}\left[(x^{(i)} - \bar{x})\theta - (y^{(i)} - \bar{y}) \right]^2 + \lambda\|\theta\|_2^2$

$= \sum_{i=1}^{m}\left[x_c^{(i)}\theta - y_c^{(i)} \right]^2 + \lambda\|\theta\|_2^2$

(C) For convenience, use matrix format.

$J_c(\theta) = (X_c\theta - Y_c)^T(X_c\theta - Y_c) + \lambda\theta^T\theta$

let $\frac{\partial J_c}{\partial \theta} = 2X_c^T(X_c\theta - Y_c) + 2\lambda\theta = 0 \Rightarrow (X_c^T X_c + \lambda I)\theta = X_c^T Y_c$, here I is a identity matrix so that $I\theta = \theta$

$\therefore \hat{\theta} = (X_c^T X_c + \lambda I)^{-1} X_c^T Y_c$

Q3

(a) $P(\hat{y} = 1 \mid x_1, x_2) = \frac{1}{1+e^{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)}} = \frac{1}{1+e^{-(-1 + 1.5x_1 + 0.5x_2)}}$

$E(\theta) = -\sum_{i=1}^{m}\left[y^{(i)} \ln\sigma(f^{(i)}) + (1-y^{(i)})(1 - \ln\sigma(f^{(i)})) \right]$, here $m=6$. In order to easily read, no data pluging here

(b) gradiant: $\frac{\partial E(\theta)}{\partial \theta_j} = \sum_{i=1}^{6}(\sigma^{(i)} - y^{(i)})x_j^{(i)}$

After calculation: $\frac{\partial E(\theta)}{\partial \theta_0} \approx -0.17004 \quad \frac{\partial E(\theta)}{\partial \theta_1} \approx -0.26984 \quad \frac{\partial E(\theta)}{\partial \theta_2} \approx -0.48935$

$\because \theta_j \leftarrow \theta_j - \eta\frac{\partial E(\theta)}{\partial \theta_j} \quad \therefore \theta_0 \approx -0.98295 \quad \theta_1 \approx 1.52698 \quad \theta_2 \approx 0.54893$

$\therefore$ new model: $\sigma(f_\theta(x_1, x_2)) = \frac{1}{1+e^{-(-0.98295 + 1.52698x_1 + 0.54893x_2)}}$

(C) After calculation: $TP = 3$, $FN = 1$, $FP = 0$, $TN = 2$

$\therefore Acc = \frac{T}{ALL} = \frac{5}{6} \quad Pre = \frac{TP}{P} = \frac{3}{3} = 1 \quad Rec = \frac{TP}{TP+FN} = \frac{3}{4}$

Q4

(a) Assume $u = X^T\theta$

I. $j = i$, then $\frac{\partial p_i}{\partial u_i} = \frac{e^{u_i}\sum_k e^{u_k} - e^{u_i}e^{u_j}}{(\sum_k e^{u_k})^2} = \frac{e^{u_i}}{\sum_k e^{u_k}} - \frac{e^{u_i}}{\sum_k e^{u_k}} \cdot \frac{e^{u_i}}{\sum_k e^{u_k}} = P(\hat{y}_i = 1 \mid x) - P(\hat{y}_i = 1 \mid x)P(\hat{y}_j = 1 \mid x)$

$= P(\hat{y}_i = 1 \mid x)(1 - P(\hat{y}_j = 1 \mid x))$

II. $j \neq i$, then $\frac{\partial p_i}{\partial u_i} = -\frac{e^{u_i}e^{u_j}}{(\sum_k e^{u_k})^2} = -\frac{e^{u_i}}{\sum_k e^{u_k}} \cdot \frac{e^{u_j}}{\sum_k e^{u_k}} = -P(\hat{y}_i = 1 \mid x)P(\hat{y}_j = 1 \mid x)$

$\therefore \frac{\partial P(\hat{y}_i = 1 \mid x)}{\partial x^T\theta_j} = \begin{cases} P(\hat{y}_i = 1 \mid x)(1 - P(\hat{y}_j = 1 \mid x)), & j = i \\ P(\hat{y}_i = 1 \mid x)P(\hat{y}_j = 1 \mid x), & j \neq i \end{cases}$

(b) $\because \frac{\partial u_i}{\partial \theta_i} = X$ and $\frac{\partial p_i}{\partial \theta_j} = \frac{\partial p_i}{\partial u_i} \cdot \frac{\partial u_i}{\partial \theta_j} = \frac{\partial p_i}{\partial u} \cdot X$

We have calculated $\frac{\partial p_i}{\partial u}$ in part (a)

$$\therefore \frac{\partial p(\hat{y}_i=1|x)}{\partial \theta_j} = \begin{cases} X\, p(\hat{y}_i=1|x)(1-p(\hat{y}_j=1|x)), & j=i \\ -X\, p(\hat{y}_i=1|x)\, p(\hat{y}_j=1=1|x), & j\neq i \end{cases}$$

(c) $\frac{\partial E(\theta)}{\partial \theta_j} = -\sum_k p(\hat{y}_i=1|x)^{-1} \frac{\partial p(\hat{y}_i=1|x)}{\partial \theta_j} y_i$

We have calculated $\frac{\partial p(\hat{y}_i=1|x)}{\partial \theta_j}$ in part (b)

$\therefore$ I. $j=i$, $\frac{\partial E(\theta)}{\partial \theta_j} = -X(1-p(\hat{y}_j=1|x))y_i$

II. $j\neq i$, $\frac{\partial E(\theta)}{\partial \theta_j} = \sum_{j\neq i} X\, p(\hat{y}_j=1|x) y_i$

$$\therefore \frac{\partial E(\theta)}{\partial \theta_j} = -X(1-p(\hat{y}_j=1|x))y_{j=i} + \sum_{j\neq i} X\, p(\hat{y}_j=1|x) y_i$$

$$= -X y_i + X\, p(\hat{y}_j=1|x) y_{j=i} + \sum_{j\neq i} X\, p(\hat{y}_j=1|x) y_i$$

$$= -X y_i + X\, p(\hat{y}_j=1|x)\left(y_{j=i} + \sum_{j\neq i} y_i\right)$$

$\because y_{j=i} + \sum_{j\neq i} y_i = 1$

$$\therefore \frac{\partial E(\theta)}{\partial \theta_j} = -X y_i + X\, p(\hat{y}_j=1|x)$$

$$-\sum_k y_i \frac{1}{R_k} \cdot \begin{matrix} R_i(1-p_j) \\ -p_i p_j \end{matrix} X$$

$$-\sum y_i (1-p_j) X$$

$$\sum_{j=i} y_i p_j X - \sum_{j=i} y_i X + \sum_{j\neq i} y_i p_j X$$

$$\sum_k y_i p_j X - \sum_{j\neq i} y_i X$$

CSCI3230 / ESTR3108 2026-27 First Term Assignment 1

I declare that the assignment here submitted is original except for source material explicitly acknowledged, and that the same or closely related material has not been previously submitted for another course. I also acknowledge that I am aware of University policy and regulations on honesty in academic work, and of the disciplinary guidelines and procedures applicable to breaches of such policy and regulations, as contained in the following websites.

University Guideline on Academic Honesty:
http://www.cuhk.edu.hk/policy/academichonesty/
Faculty of Engineering Guidelines to Academic Honesty:
http://www.erg.cuhk.edu.hk/erg-intra/upload/documents/ENGG_Discipline.pdf

Student Name: Yuzhe ZHANG
Student ID : 1155211114